CLINICAL · LARYNX & AIRWAY

Hoarseness, voice and speech disorders

A voice is made by three things working together: the lungs as a power supply, the vocal folds as a vibrator, and the throat, mouth and nose as resonators and articulators. Hoarseness is a disturbance of the vibrator; nasal speech a disturbance of the resonator; stuttering a disturbance of the rhythm. Most voice complaints are benign, many are functional, and a few are the first sign of cancer or neurological disease. Look at every voice that has been wrong for more than a few weeks, and listen to it as carefully as you look.

AT A GLANCE

- Functional aphonia: folds close on coughing.
- Puberphonia: Gutzmann pressure test.
- Spasmodic dysphonia: a dystonia; botulinum toxin.
- Hyponasality is blockage; hypernasality is velopharyngeal insufficiency.
- "Spasmodic dysphonia" is now laryngeal dystonia; inducible laryngeal obstruction often coexists with chronic cough.

1 CHAPTER ONE Hoarseness

The commonest voice complaint.

1.1 Definition

Hoarseness (dysphonia) is roughness of voice from irregularity of the period or intensity of successive sound waves. Normal voice needs folds that (1) approximate properly, (2) have the right size, mass and stiffness, and (3) vibrate regularly in the air stream.

- **Failed approximation:** paralysis, fixation, a mass between the folds, sulcus, bowing.
- **Altered size and mass:** oedema, nodules, polyps, tumours (more); surgical loss or fibrosis (less).
- **Altered stiffness:** paralysis (less), scar and spasm (more).
- **Irregular vibration:** congestion, haemorrhage, lesions on the edge.

1.2 Causes

CAUSES OF HOARSENESS

GROUP	EXAMPLES
Inflammation	Acute viral laryngitis, diphtheria, irritant gases; chronic laryngitis from smoking, reflux, occupation and inhaled steroids; tuberculosis, syphilis, leprosy, fungi
Benign lesions	Nodules, polyp, Reinke's oedema, cyst, contact granuloma, laryngocele, amyloid, sulcus
Neoplasms	Papilloma, haemangioma, chondroma, granular cell tumour; leukoplakia; carcinoma and sarcoma
Trauma	Shouting with fold haemorrhage, external injury, foreign body, intubation
Neurological	Recurrent, superior laryngeal or combined paralysis; spasmodic dysphonia; Parkinson disease, myasthenia, motor neurone disease, multiple sclerosis, essential tremor
Joint fixation	Cricoarytenoid arthritis, dislocation
Congenital	Web, cyst, laryngocele, paralysis, sulcus
Systemic and hormonal	Hypothyroidism, acromegaly, virilising hormones, sarcoidosis, granulomatosis with polyangiitis, amyloidosis
Functional	Muscle tension dysphonia, ventricular dysphonia, functional aphonia, puberphonia, phonasthenia
Ageing	Presbyphonia with bowed, thin folds

1.3 Evaluation

1. **History:** onset and duration, voice use and occupation, smoking, alcohol, reflux, inhalers, systemic symptoms, surgery (thyroid, chest, neck), and red flags (pain, otalgia, dysphagia, stridor, haemoptysis, weight loss, neck mass).
2. **Listening:** breathy (incomplete closure), rough (irregular vibration), strained (hyperfunction or spasm), weak, diplophonic, tremulous.
3. **Laryngoscopy:** any hoarseness lasting more than two to three weeks, and sooner in a smoker or with red flags, needs the larynx examined; malignancy must be excluded over forty.

4. **Stroboscopy:** the mucosal wave, closure pattern and symmetry.
5. **Neck, chest, cardiovascular and neurological examination** for the cause of a paralysis; imaging, endoscopy and laboratory tests as indicated.
6. **Direct and microlaryngoscopy** for biopsy, palpation of the arytenoids and treatment.
7. **Voice measures:** maximum phonation time, acoustic analysis, the GRBAS rating, and patient-reported voice handicap indices.

CHAPTER TWO

2 Functional voice disorders

Normal larynx, abnormal use.

2.1 Muscle tension dysphonia

Excessive tension of the laryngeal and neck muscles during phonation: strained voice, vocal fatigue and neck ache, with supraglottic squeezing, a posterior gap or anteroposterior compression on laryngoscopy. It may be primary or compensatory. Voice therapy with laryngeal manual therapy is the treatment.

2.2 Ventricular dysphonia (dysphonia plicae ventricularis)

The false folds take over phonation: a rough, low-pitched, unpleasant voice; on phonation the false folds meet and hide the true folds.

- **Secondary:** compensating for true fold paralysis, fixation, excision or tumour; difficult to treat.
- **Functional:** in a normal larynx, often psychogenic; the voice starts normally and then roughens. Voice therapy and counselling.

2.3 Functional (psychogenic) aphonia

Sudden loss of voice, usually in emotionally stressed young women of fifteen to thirty, communicating in a whisper without other laryngeal symptoms. The folds stay apart on attempted phonation but *close normally on coughing*, and the cough is loud and clear. Reassure the patient that the larynx is normal, restore voice through

coughing, humming and throat clearing exercises, and address the stress; psychotherapy as needed.

2.4 Puberphonia (mutational falsetto)

Persistence of the high-pitched childhood voice in a physically normal adolescent boy or young man, often from emotional immaturity or social insecurity, while the larynx has matured. *Gutzmann's pressure test*: pressing the thyroid prominence backwards and downwards relaxes the overstretched folds and produces a low-pitched voice; the patient learns to reproduce it, first with pressure, then without, through syllables, words and sentences. The prognosis is good; type III thyroplasty is a last resort.

2.5 Phonasthenia

Weakness and fatigue of the phonatory muscles after misuse or laryngitis: a voice that tires quickly. Laryngoscopy shows an elliptical gap between the folds (thyroarytenoid weakness), a triangular posterior gap (interarytenoid weakness), or a *keyhole* glottis (both). Voice rest, vocal hygiene and therapy.

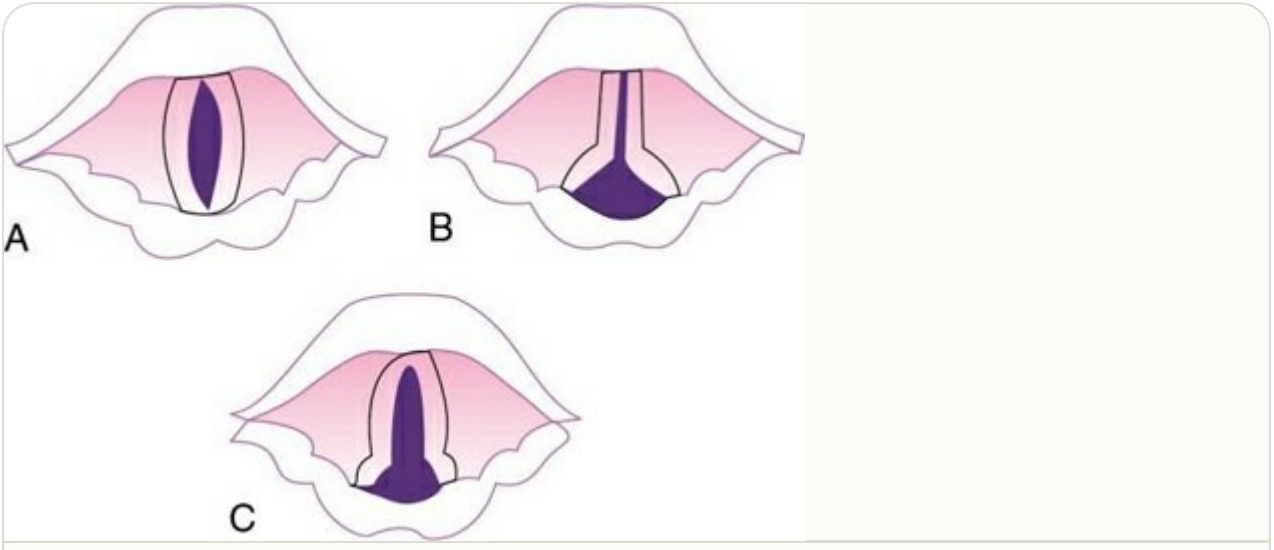


FIGURE 1 The glottis in phonasthenia: (A) weakness of the thyroarytenoids, (B) of the interarytenoid, (C) of both, giving a keyhole glottis.
Dhingra, fig. 63.1

3 CHAPTER THREE Spasmodic dysphonia

A laryngeal dystonia.

3.1 Nature

A focal dystonia, thought to arise in the basal ganglia, with involuntary spasms of the laryngeal muscles during speech; the larynx is structurally normal and behaves normally for breathing, laughing, crying, whispering and often singing.

3.2 Types

- **Adductor (about 90%):** spasms of the thyroarytenoid and lateral cricoarytenoid; a strained, strangled, effortful voice with breaks, especially on words beginning with vowels.
- **Abductor:** spasms of the posterior cricoarytenoid; a breathy voice with breathy breaks after voiceless consonants (p, t, k), worse with stress and on the telephone.
- **Mixed.**

Voice tremor often coexists. Exclude Parkinson disease, essential tremor, multiple sclerosis, motor neurone disease and tardive dyskinesia.

3.3 Treatment

- **Botulinum toxin A** is the treatment of choice: small doses into the thyroarytenoid–lateral cricoarytenoid complex on one or both sides for adductor spasm, through the cricothyroid membrane under EMG guidance or through a flexible scope. Benefit lasts about three to four months; side effects are temporary breathiness and choking on fluids. For abductor spasm, injection into the posterior cricoarytenoid is harder and less effective, and is staged one side at a time to avoid airway obstruction.
- **Voice therapy** helps alongside toxin but not alone; psychotherapy is ineffective.
- **Surgery:** recurrent nerve section (abandoned because symptoms return); selective adductor denervation–reinnervation; type II (lateralisation) thyroplasty; thyroarytenoid myectomy.

OPERATIVE GUIDES IN THEATRE 1 operation

Office-based laryngeal procedures, including botulinum toxin injection.

Office-Based Laryngeal Procedures
Office procedures and botulinum toxin

MOVEMENT DISORDERS OF THE LARYNX (OXFORD TEXTBOOK)

Laryngeal dystonia is the term a recent multidisciplinary consensus proposed to replace "spasmodic dysphonia". It is task-specific: speaking is affected, but whispering, laughing and crying are not. *Adductor* dystonia, the commonest, gives strained-strangled breaks on voiced sounds; *abductor* dystonia gives breathy breaks on voiceless consonants; mixed forms occur. Newer variants are *singer's* dystonia, affecting only singing, and *adductor respiratory* dystonia, with adductor spasms on inspiration causing stridor. A simple bedside test is to count from 80 to 90 (voiced sounds, worse in adductor dystonia) and from 60 to 70 (voiceless sounds, worse in abductor dystonia).

Vocal tremor may be isolated or part of essential tremor of the head and hands, and is seen as rhythmic movement of the larynx and pharynx at rest phonation. **Parkinson's disease** gives a quiet, breathy, monotonous voice with bursts of rapid speech, usually after the limb signs; intensive loudness-focused voice therapy helps.

Inducible laryngeal obstruction (formerly paradoxical vocal fold movement or vocal cord dysfunction) is inappropriate adduction of the folds on inspiration, triggered by exercise, irritants or cough, and often mistaken for asthma. Laryngoscopy is usually normal unless done during symptoms or provocation, and many patients have sensations of tightness and altered breathing rather than visible closure. It often coexists with **chronic refractory cough** (cough over eight weeks, affecting 5 to 10% of adults), and both are best managed in a multidisciplinary upper airway clinic with breathing and laryngeal control techniques taught by speech and language therapists.

4 CHAPTER FOUR Resonance and fluency

Beyond the larynx.

4.1 Hyponasality (rhinolalia clausa)

Lack of nasal resonance on the nasal consonants m, n and ng ("my nose" sounds like "by dose"), from blockage of the nose or nasopharynx.

4.2 Hypernasality (rhinolalia aperta)

Too much nasal resonance on oral sounds, with nasal escape, because the nasopharynx is not sealed off from the oropharynx or there is a communication between mouth and nose.

CAUSES OF ALTERED NASAL RESONANCE

HYPONASALITY	HYPERNASALITY
Common cold, allergic rhinitis	Velopharyngeal insufficiency
Nasal polyps and tumours	Cleft palate and submucous cleft
Adenoids, nasopharyngeal mass	Congenitally short palate, large nasopharynx
Choanal atresia	Palatal paralysis
Habitual or familial	After adenoidectomy, oronasal fistula; habitual or familial

Hypernasality is assessed by perceptual rating, nasometry, nasendoscopy and videofluoroscopy of velopharyngeal closure, and treated by speech therapy, prosthetic palatal lifts, or surgery (pharyngeal flap, sphincter pharyngoplasty, palate repair, posterior wall augmentation).

OPERATIVE GUIDES IN THEATRE 1 operation

Cleft palate and velopharyngeal dysfunction.

Cleft Palate and Velopharyngeal Dysfunction
Velopharyngeal dysfunction

4.3 Stuttering

A disorder of fluency: hesitation, repetitions, prolongations and blocks, with secondary mannerisms such as grimacing, blinking and head jerks. Normal dysfluency is common between two and four years; parental overreaction may fix it. Prevention by educating parents; treatment by speech therapy and psychological support to reduce the fear of speaking.

4.4 Dysarthria and aphasia

Dysarthria is a motor disorder of speech production (flaccid, spastic, ataxic, hypokinetic, hyperkinetic); aphasia is a disorder of language (expressive, receptive). Both are neurological and need neurological assessment.

5 CHAPTER FIVE Voice care

Prevention and therapy.

VOCAL HYGIENE

- Hydrate well; humidify dry air.
- Avoid smoking, excess alcohol and caffeine; treat reflux.
- Avoid shouting, prolonged loud talking, whispering when hoarse, and habitual throat clearing.
- Rest the voice after heavy use; warm up before performing.
- Use amplification in noisy rooms and classrooms.

Voice therapy by a speech and language therapist retrains breathing, onset, pitch, resonance and muscle tension; it is the first treatment for nodules, muscle tension dysphonia, functional disorders and many benign lesions, and supports surgery.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Definition and causes** of hoarseness, and the two-to-three-week rule.
- **Functional aphonia:** the folds close on coughing.
- **Puberphonia** and Gutzmann's test.
- **Phonasthenia:** elliptical, triangular and keyhole glottis.
- **Spasmodic dysphonia:** types and botulinum toxin.
- **Hyponasality versus hypernasality** with causes.

6.2 Common pitfalls

COMMON PITFALLS

- Diagnosing functional aphonia without seeing the larynx.
- Calling spasmodic dysphonia psychogenic.
- Removing adenoids in a child with a submucous cleft and causing hypernasality.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, 72, Oxford Specialist Handbook ch. 8, 13, Oxford Handbook of ENT ch. 22, Dhingra ch. 63 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

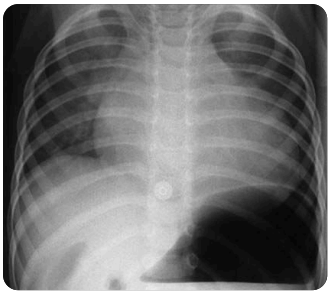
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.